

MD NAFIZUR NAYEM

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SUMMARY

Final-year CSE student at American International University - Bangladesh (AIUB) with hands-on experience building production-grade AI, IoT, and full-stack systems. Published researcher in autonomous robotics with 7+ end-to-end projects spanning Gemini AI integration, ESP32 firmware, Flutter apps, and Next.js platforms. Proven ability to lead technical teams and deliver cross-disciplinary engineering solutions from hardware to cloud.

EDUCATION

American International University-Bangladesh (AIUB) <i>Bachelor of Science in Computer Science and Engineering</i>	Sep 2022 – Present CGPA: 3.14 / 4.00
Govt. Haraganga College <i>Higher Secondary Certificate (HSC)</i>	Jan 2021 – Feb 2022 GPA: 5.00 / 5.00
President Prof. Dr. Iajuddin Ahmed Residential Model School & College <i>Secondary School Certificate (SSC)</i>	Jan 2019 – Dec 2020 GPA: 4.83 / 5.00

SKILLS

Programming Languages: Python, C, C++, C#, Java, JavaScript, TypeScript, SQL, R, PHP
AI & Machine Learning: Supervised & Unsupervised Learning, NLP, LLM APIs (Gemini), Computer Vision, Agentic AI, scikit-learn
Web & Mobile: HTML, CSS, Next.js, FastAPI, Streamlit, Flutter, REST APIs, .NET / Windows Forms
Embedded & IoT: ESP32, ESP8266, Arduino UNO, Raspberry Pi 5, MQTT (HiveMQ), Home Assistant
Databases & DevOps: MySQL, Oracle, Docker, Git, Linux, Google Colab, Kaggle, Vercel
Design & Office: Adobe Photoshop, Illustrator, Audition, Microsoft Office Suite
Soft Skills: Technical Leadership, Cross-functional Collaboration, Research, Adaptability, Volunteering

PROFESSIONAL EXPERIENCE

AIUB Community of Engineering Students (ACES) <i>IT & Social Media Coordinator</i>	Jan 2024 – Aug 2026 Dhaka, Bangladesh
- Promoted from Volunteer to Coordinator within 12 months by demonstrating exceptional work quality, leading a team of 5+ volunteers across IT operations and digital content for a 150+ member engineering community. - Spearheaded digital strategy across 3 platforms, coordinating 8+ technical seminars and workshops that measurably increased community event attendance and online engagement. - Planned and executed 5+ high-impact community events end-to-end, managing cross-functional volunteer teams from logistics and technical setup through to post-event reporting.	

PROJECTS

Smart Lamp System Python, Gemini Vision API, OpenCV GitHub	2026
- Built a Gemini-powered educational smart lamp with an integrated camera and touchscreen display, capturing and digitizing multi-page handwritten notes in real time via finger gesture recognition. - Leveraged Gemini Vision API to automatically evaluate math equations, correct grammar and spelling, and export processed content to Word or PDF — reducing manual transcription effort to near zero.	
E-Commerce Platform (EverShop) Next.js 15, TypeScript, Vercel Live GitHub	2025

- Developed a full-featured e-commerce platform with separate admin and user panels, real-time inventory tracking, and payment integration tailored for the Bangladeshi market with BDT currency support.
- Implemented authentication, dynamic product management, and order processing using Next.js 15 server actions; deployed to Vercel with a zero-downtime CI/CD pipeline.

Mini PC Assistant | ESP32-S3, Flutter, .NET, MQTT, Home Assistant | [GitHub](#)

2026

- Engineered a cross-platform IoT PC control system spanning ESP32-S3 firmware, a Flutter mobile app, and a .NET Windows background service — enabling secure remote wake, shutdown, and real-time telemetry monitoring.
- Integrated HiveMQ MQTT broker and Home Assistant for smart-home automation, achieving sub-2 second command latency over mobile data networks.

Smart Solar Power Management | ESP32, C++, Embedded Web App | [GitHub](#)

2024

- Designed a low-cost solar energy sharing system supporting 2 user roles (Prosumer and Consumer) with configurable pricing, usage limits, and real-time electrical measurement via onboard sensors and live display.
- Built an embedded web interface for remote monitoring and configuration, eliminating the need for a dedicated server and significantly reducing system deployment cost vs. commercial alternatives.

Smart Home Automation System | ESP32, ESP RainMaker, Google Home | [GitHub](#)

2024

- Built an ESP32-based smart home system controlling 4+ devices (lights, fans) via Google Home, IR remote, and manual switches, with real-time DHT11 temperature and humidity monitoring.
- Used ESP RainMaker cloud for OTA-configurable device provisioning, enabling remote control from anywhere without requiring a local server.

Restaurant Management System | C#, .NET Framework, Windows Forms | [GitHub](#)

2023

- Developed a desktop application handling staff management, inventory tracking, table reservations, and daily sales reporting for restaurant operations with a role-based access control system.

Power On PC Remotely | ESP8266, Arduino, Blynk IoT | [GitHub](#)

2023

- Designed an ESP8266-based Wake-on-LAN solution using Blynk cloud to remotely power on a PC or laptop from anywhere via a mobile app, bypassing traditional network-restricted WoL limitations.

RESEARCH

IoT-Based Autonomous Agribot for Precision Farming | [ResearchGate](#)

2025

- Designed and implemented an IoT-enabled autonomous agricultural robot integrating real-time sensor data, autonomous navigation, and remote monitoring to improve precision farming accuracy and efficiency.
- Published peer-reviewed research on ResearchGate detailing full system architecture, field test results, and performance benchmarks.

ADDITIONAL INFORMATION

Languages: Bangla (Native), English (Professional Proficiency)

Certifications & Achievements:

- **Digital Manufacturing Certification** — Awarded jointly by AIUB and Birmingham City University (UK) under British Council funding, covering Industry 4.0 and smart manufacturing systems. ([Certificate](#))
- **IT Essentials: PC Hardware & Software** — Cisco Networking Academy verified certification covering hardware troubleshooting, networking fundamentals, and OS administration. ([Certificate](#))
- **Line Follower Robot — National Innovation Competition 2024 (Technocrats V.2, IUBAT)** — Competed in autonomous robotics, applying embedded systems and real-time control algorithms. ([Certificate](#))
- **Line Follower Robot — BITFEST 2025 (KUET)** — Participated in the robotics competition organized by the CSE Department of Khulna University of Engineering & Technology. ([Certificate](#))